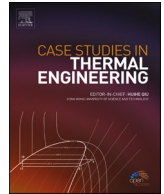




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Feasibility of integrated photovoltaic and mechanical storage systems for irrigation purposes in remote areas: Optimization, energy management, and multicriteria decision-making

Bashri A.A. Yousef^{a,b,*}, Rahaf Amjad^a, Noura Ali Alajmi^a, Hegazy Rezk^c

^a University of Sharjah, College of Engineering, Sustainable and Renewable Energy Engineering Department, Sharjah, P. O. Box 27272, United Arab Emirates

^b Department of Mechanical Engineering, Bahri University, North Khartoum, Khartoum, Sudan

^c Department of Electrical Engineering, College of Engineering in Wadi Alldawasir, Prince Sattam bin Abdulaziz University, Saudi Arabia

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ABSTRACT

Remote and rural areas are suffering from the lack of electricity, which makes electrical power needed for irrigated pumps one of the challenges in those regions. This paper proposed a hybrid system consisting of photovoltaic and different sizes of diesel generators as the main energy production source, flywheel, and batteries as storage systems. HOMER software is used to determine the optimum configuration of the proposed system, as well as to analyze its economic and environmental impact. The investigation was conducted on twelve alternatives that include different energy management strategies and sizes of diesel generators. Seven criterions involving technical, economic, and environmental metrics are considered during the assessment process. Multicriteria decision-making (MCDM) based on; no priority, CRITIC and Entropy as different approaches of weight has been applied to identify the best alternative. Alternative A8 which includes 750 kW DG size and combined LF–CC energy management is ranked as the first, followed by alternative A4 which comprises 500 kW size of DG and applies LF management strategy. Alternative A11 which includes DG of 750 kW and predictive control energy management is ranked as the worst one. The case study was conducted and simulated based on the weather of Atbara, Sudan.

1. Introduction

The concern with renewable energy has been grown recently as an alternative to fossil fuels, due to the pollution released during the use of fossil fuels. Fossil fuels include coal, oil, and natural gas, however, coal is the most polluted type of fossil fuel, it is liable for 0.3 °C of the 1 °C increase in global average temperatures, which makes it the largest source of global temperature rise [1,2]. The energy sector is a vital domain, and with the world population being aware of the dangers caused using fossil fuels, the market has increased in demand for clean energy. Consequently, researches and applications of renewable energy have grown to fulfil the demand required in the market [3].

One area of study that is still largely dependent on fossil fuels as a means for energy production is irrigation. Irrigated land typically

* Corresponding author. University of Sharjah, College of Engineering, Sustainable and Renewable Energy Engineering Department, Sharjah, P. O. Box 27272, United Arab Emirates.

E-mail address: byousef@sharjah.ac.ae (B.A.A. Yousef).

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contains crops, grass, or vegetation will not receive enough water from rainfall or other natural sources if it is not irrigated. In remote and rural areas the electrical pump is considered costly and challenging procedure, thus pumps and other electrical devices in remote areas are usually powered by diesel generators. One of the solutions to decrease the use of diesel generators is to incorporate or completely replace the energy production unit with renewable energy [4]. In this content, Shahverdi et al., investigated the possibility of supplying power required by pressurized irrigation system via solar driven Organic Rankine Cycle (ORC) (Fig. 1). The parabolic trough and thermal energy storage were the heat source for ORC that produce the required electricity for pumping system [5].

A stand-alone solar PV battery irrigation system was studied by Rezk et al. for the purpose of evaluating performance and determining optimal design. The authors compared the cost of energy of their study with previously reported studies. They found that it is lower than the previously reported studies due to the appropriate selection of PV size, type, and location. They found that an amount of 86,511 kg/year of CO₂ emissions is reduced for PV/BS compared to DG system [6]. Purohit [7], developed a simple framework to evaluate multiple renewable energy technologies including photovoltaic, windmill, biogas and dual-fuel engine used to operate pumps for irrigation. The author reported that the dual fuel engine pumps using biogas are the best cost-effective choice for water pumping, with the unit cost of high capacity biogas plants being further reduced due to the effect of economies of scale in their capital costs [7].

When a system incorporates one or more sources of energy as the energy production source, the system is said to be a hybrid system. Because these systems usually integrate more than one energy source, they need energy management and optimization. Several authors investigated hybrid systems for different applications. The performance of hybrid photovoltaic thermal solar collector system that provide hot water, space heating and electrical loads for a home with an area of 100 m² is experimentally investigated by Ref. [8] for Duhok city/Iraq. The study showed that panel temperature has a significant effect on the electrical production, the thermal and electrical energy productions are proportional with the insolation. However, the produced thermal energy is triple the electrical energy production. A hybrid system consists of solar PV and wind turbine power system is investigated and optimized by Ghenai et al., [9]. The hybrid system is designed and tested to supply clean energy demand of a pyrolysis reactor. Moreover, the modelling and optimization analysis is performed to simulate the off-grid renewable power system. The experimental results showed that the proposed hybrid power system met the energy demand of the reactor. On the other hand, the simulation results showed that the off-grid hybrid power system can increase renewable fraction and reduce the CO₂ emissions, but the with high cost of electricity.

Jaber et al. [10], proposed a new hybrid heat recovery system that utilizes exhaust gases of a generator to heat water and produce electricity using thermoelectric generators. The authors studied the effect of changing the load of the generator on the water temperature and power generated. The results showed that water temperature and generated power are increased as the load of the generator is increased. Obaid et al. [11], proposed a hybrid renewable and conventional power system for water supply applications in Dubai. The hybrid system consists of solar panels and turbines as a renewable power system part, and diesel generator as a conventional power system part. DC voltage from the solar power system stabilized by regulator and stored in the battery bank. The voltage of wind turbine converted to DC via three-phase rectifier. The proposed system demonstrated with SIMULINK and MATLAB. The feasibility of a hybrid renewable energy system for irrigating a small farm in Egypt has been studied also, by Khattab et al. using commercial software, the article theoretically assessed various renewable energy water pumping systems with various load patterns [12].

The use of a storage device with photovoltaics (PV) energy production is necessary, as PV technology is an intermediate power source and needs a storage device, to store the excess energy produced and use it when needed. Photovoltaic, fuel cell and hydrogen storage have been integrated to make a stand-alone Poly-generation Micro-grids (PMG) system for decentralized multi-output. The system has been modelled and optimized using HOMER Software and a case study were conducted for typical Indian village that required 100 kWh/day. Through the study, the optimized sizes of hybrid system components needed to meet the load demand have been evaluated [13]. Keeping in mind the importance of battery storage system for PMG the authors investigated in another study the influence of battery storage on the performance [14]. The analysis and performance of integrated PV and fuel cell with diesel generator and different types of storage system were presented by Ref. [15]. The authors used HOMER Pro software to model the integrated energy system. The system was modelled based on the daily energy consumption for Khorfakkan city in United Arab Emirates (UAE). The simulation results showed the effectiveness of using energy storage system on both Levelized cost of energy (LCOE) and gas

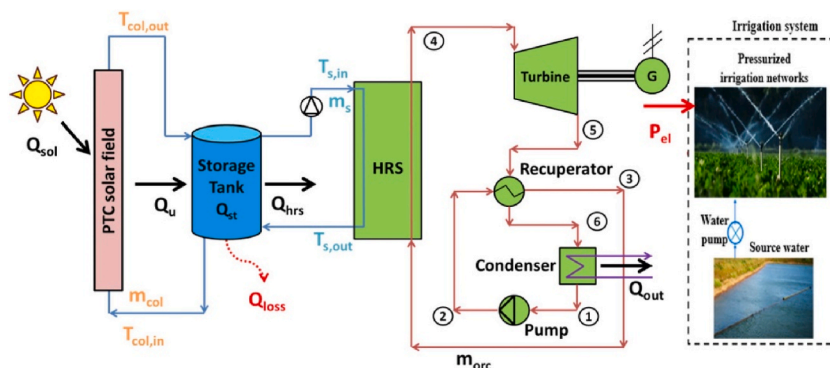


Fig. 1. Solar driven Organic Rankine Cycle for irrigation system, (licence No. 5360070530610) [5].

emission [15].

Many storage devices have been tested along with photovoltaics. The type of storage system depend on the forms they can store the energy; such as chemical, kinetic, thermal, potential, electromagnetic and electrochemical forms. They are known as fuel cells, batteries, capacitors, flywheels, pumped hydro, hydrogen, and more [16]. The most used Electrical Energy Storage (EES) devices are the batteries in spite of their negative impact on environment, the user keep using them because of their availability, low cost, and their well-known process. Therefore, it is unrealistic to remove the batteries from the system entirely. However, it is necessary to consider combining more than one energy storage device to reduce the environmental impact that batteries can cause.

Another EES system that in use is Flywheel. The usage of a flywheel provides another backup option for improving energy supply reliability and potentially reducing diesel generation. They stored energy in the form of kinetic energy, by applying the conservation of angular momentum to store the energy in the system as a rotational energy [17]. Flywheels have gained interest because of their advantages, which include a high cycle life, round-trip efficiency, power density and low maintenance, environmental impact, and rapid charge and discharge rates [18]. In term of power storage, the flywheel is competitive with batteries. The main differences between flywheels and the commonly used batteries are shown in Table 1.

The purpose of this research work is to examine the potential of employing hybrid energy system combined with storage systems for irrigation purposes in remote and rural areas. The hybrid system integrates photovoltaic (PV) and diesel generator (DG) as the primary energy source, flywheel (FW), and battery storage (BS) as the storage systems for excess energy. A case study of the proposed system is done on an irrigation farm in the State of Nile River, Atbara, Sudan. Four energy management strategies; Load Following (LF), Cycle Charging (CC), combined LF–CC, and Predictive Strategy (PS); and three different sizes of diesel generator; 500 kW, 750 kW and 1000 kW are investigated to select the best alternative for the case study. Seven criterions including initial cost (\$), operating cost (\$/yr), net present cost (\$), COE (\$/kWh), renewable fraction (RF) (%), excess energy (%) and CO₂ emissions (kg/y) are used to assess the different alternatives.

The process of directly determining the optimal energy management strategy and size of diesel generator is challenging, since none of the assessed alternatives has the best parameters among the seven criterions. Therefore, and to solve this dilemma, Multi-Criteria Decision-Making (MCDM), based on a different approach of weight namely, no priority, CRITIC, Entropy have been applied. Additionally, to select the best alternative more three methods of MCDM namely, technique for the order of preference by similarity to the ideal solution (TOPSIS), weighted aggregated sum product assessment (WASPAS), and weighted sum and weighted product methods (WSM/WPM) were applied. Up to the knowledge of the authors, it is for the first time the MCDM based on TOPSIS, WASPAS and WSMWPM techniques along with no priority, CRITIC, and Entropy approaches used to determine the optimal energy management strategy and size of diesel generator simultaneously.

2. Configuration of the proposed system

The model in Fig. 2 was developed using HOMER, which is developed by the National Renewable Energy Laboratories (NREL) and used for micro-grid system design and analysis. The software was used to conduct an economic study of a hybrid PV/diesel system with different energy storage systems. The components of the system comprised of diesel generator, PV module, chemical and mechanics storage systems (batteries and flywheel respectively). The detail description of each component is presented in the following section, and the data components is listed in Table 2.

Table 1
The differences between batteries and flywheels [19–22],

Parameter	Batteries	Flywheels	Remarks
Storage type	Chemical	Mechanical	Batteries are flammable, contain materials that can be toxic to the environment, and are difficult to recycle. The flywheel uses recycled steel that is recyclable & non-toxic.
Efficiency	50%–95%	90%–95%	Both have relatively good round-trip efficiency.
Operating temperature.	–35 °C–45 °C	–40 °C–70 °C	Flywheels are insensitive to temperature while batteries cannot be charged at temperatures approaching (–20 °C) and have low efficiency at high temperatures.
Cycle life	<350 cycles	>50,000 cycles	The battery cycle is dependent on how much the battery is discharged in each cycle, unlike the flywheel cycle, which is not affected by the depth of discharge.
Design life	3–5 years	25 years	This will add to the cost of the system, as batteries will have to be replaced at least twice.
Energy density	35–40 Wh/kg	27–43 Wh/kg	Both have dependable energy density value.
Weight	Light	About 5 times heavier	Flywheels are heavy and not practical for vehicles.
Volume	small	small	Both batteries and flywheels occupy a relatively small volume
Levelized cost of storage (LCOS)	11c/kWh	3.8c/kWh	Levelized Cost of Storage (LCOS) for 25-year life micro-grid applications at 1 cycle/day.
Infrastructure	Requires separate building with conditioned space	Standard blockhouse or below grade sewer pipelined shaft	This will add to the total cost of the project.

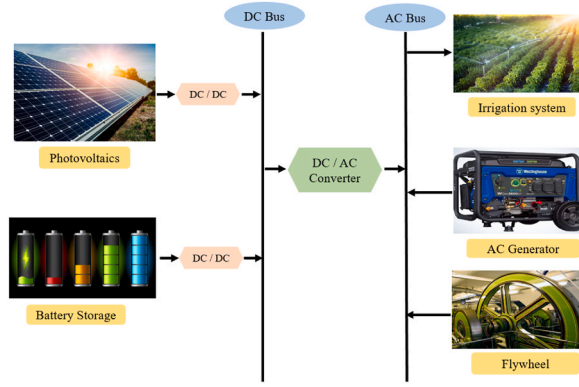


Fig. 2. Schematic of the proposed system.

Table 2

List of the data of all systems components.

Diesel generator Data	
Initial capital	\$ 150,000
Replacement cost	\$ 150,000
Operation and maintenance cost	5 \$/h
Fuel price	1.09 \$/L
Lifetime	15,000 h
Min. Load ratio	25%
PV module Data	
Initial capital	\$ 640/kW
Replacement cost	\$ 640/kW
Operation and maintenance cost	\$ 10/yr
Lifetime	30 yr
Derating factor	80%
Battery Data	
Initial capital	\$ 1200/quantity
Replacement cost	\$ 1200/quantity
Operation and maintenance cost	\$ 60/yr
Lifetime	12 yr
Batteries per string	12
Capacity	1350 Ah
voltage	4 V (in total 48 V)
Flywheel data	
Initial capital	\$ 400,000/quantity
Replacement cost	\$ 200,000/quantity
Operation and maintenance cost	\$ 8000/yr
Lifetime	20 yr
Charge/discharge capacity	500 kW
Inverter data	
Initial capital	\$ 400/kW
Replacement cost	\$ 375/kW
Operation and maintenance cost	\$ 20/yr
Lifetime	30 yr
Efficiency	90%

2.1. Diesel generator (DG)

Three diesel generators with different sizes of 500 kW, 750 kW and 100 kW were used to evaluate all possible simulations that would yield the best results. The Fuel consumption and efficiency are considered as essential parameters for DG. To estimate fuel consumption, which corresponds to the amount of fuel used in generating electricity, the formula below is used [23]:

$$F_d = (a.T_d + b.P_d) \quad (1)$$

Where F_d is the diesel generator's fuel consumption rate (L/h), T_d is the diesel generator's capacity and P_d is the diesel generator's output, a is fuel intercept coefficient (L/kWh) and b fuel slope (L/kWh).

The fuel intercept coefficient (a) and fuel slope (b) for DG are derived using its performance data and fuel consumption curve provided by HOMER, yielding $a = 0.041\text{L/kWh}$ and $b = 0.244\text{L/kWh}$ respectively.

2.2. PV module

Usually, the amount of power produced by PV module is based on its size. In order to estimate the size of a PV module, the total peak watt produced has to be known, which depends on the size of the PV module and climate of site location [24]. Therefore, according to the amount of unmet electric load and renewable portion selected in the design, PV size either can increase in number (over-sized) or decrease in number (under-sized).

The output power from the PV panel always depends on the temperature as shown in the relation below [25].

$$P_{PV} = f_{PV} \times Y_{PV} \times \frac{G_T}{G_S} \times \{1 + a(T_c - T_{c,ref})\} \quad (2)$$

Where is P_{PV} and Y_{PV} are the electrical and maximum power output from PV respectively, G_T and G_S are actual and standard solar irradiance respectively, T_c and $T_{c,ref}$ are PV module and reference temperature, a , is the temperature coefficient of power and f_{PV} is derating factor. The PV module used in this study is Peimar SG200M5 with a rated capacity of 200 kW and an efficiency of 15.7%.

2.3. Batteries

As shown in Fig. 3, throughput limits the battery's life, therefore, it is assumed that the battery will need to be replaced after a certain number of energies cycles and is not dependant on the depth of single charge and discharge cycles. Moreover, the battery lifetime throughput is used to calculate the lifetime of the battery. The safe Operational State of Charge (SOC) of the battery is in the range of (40%–100%). When operated within the safe region, the battery lifetime throughput is 10,552 kWh, and its lifetime is 12 years. As shown in Fig. 3, if the battery is operated below 40% SOC or beyond 60% Depth of Discharge (DOD), the battery lifetime will be lowered. Thus, in order to sustain the lifetime of the battery, the DOD should stay relatively around 50% [23]. The chosen battery for this study is Surrette 4 KS25P type, it is a 4 V deep cycle battery rated at 1811 Ah for 100 h of operations. A 48 V string was created by connecting twelve 4-V batteries in series, while a battery bank was created by connecting strings in parallel.

2.4. Flywheel

The flywheel can reserve and transmit the electrical energy needed for a brief amount of time without needing to be recharged. It is one of the energy storage techniques that can be used with medium-scale sustainable energy sources like solar power and wind power [26]. They also have a wide range of uses in power stability and quality management, that include power changes and disturbances in wind energy systems, where they provide backup energy [27]. The storage time of a flywheel energy storage system is typically between 5 and 30 s. A flywheel charges and discharges at high speeds for many cycles. Furthermore, the electrical energy is stored in the flywheel by mechanical operations. This happens when the said electrical energy stored is transformed to the kinetic energy of the spinning mass called the rotor. The flywheel stores energy by speeding the spinning mass to a high speed and then releasing it by decreasing the speed of the rotating mass until it comes to a complete halt.

The mass and angular velocity of the rotor govern the amount of energy stored. The stored kinetic energy (E) equation is [23]:

$$E = \frac{1}{2} J \omega^2 \quad (3)$$

where ω is the angular velocity of the rotor and J is the moment of inertia, which is calculated by Ref. [23]:

$$J = k m r^2 \quad (4)$$

Where m is the rotor mass in kilograms, r is the rotor radius in meters, and k is an inertial constant that varies depending on the rotor form.

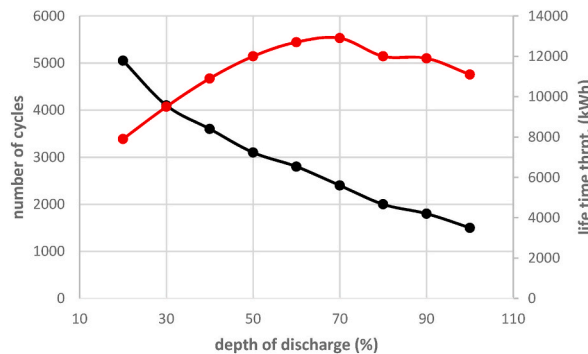


Fig. 3. Lifetime curve data extracted from Ref. [23].

The flywheel chosen for this system is the ABB PowerStore-500 flywheel which has both a charge and discharge capacity of 500 kW and an energy capacity of 18 MW. It is important to note that the flywheel should give the DGs enough time to start up. Hence, the flywheel's backup period is roughly 30 s, this allows the DG plenty of time to start up and then synchronize with the system [23].

2.5. Inverter

Initially, the inverter size is determined by the PV array size, and this is done to give the maximum amount of energy gathered by the inverter from the PV arrays. In most cases, a constant R is employed to show the PV size in relation to the inverter size, which is calculated as:

$$R = \frac{\text{the PV size}}{\text{the inverter size}} \quad (5)$$

Furthermore, the inverter is usually sized equal to or sometimes less than the PV size because the rated power produced by the PV isn't always produced in full. The decision to utilize a smaller inverter is motivated by the need to reduce inverter costs. Most PV inverters, on the other hand, are unable to generate reactive electricity. In that instant, the inverters of the PV should be larger with respect to the PV arrays so that the required reactive power may be delivered without affecting active power output. When all economic factors are taken into account, also larger inverters are said to have the best size at a power factor of one [28]. The inverter's lifetime was set to 30 years, this is the life expectancy suggested by the IEA for utility-scale inverters [29].

3. Site location and load profile

The proposed hybrid systems are investigated for a location in Sudan at State of Nile River (Latitude $17^{\circ} 42'$ Latitude N $33^{\circ} 59'$ Longitude E), in El Makaylab region that characterised by fertile agricultural land and located 22 km north Atbara city and 7 km South Berber as illustrated in Fig. 4. The irrigated area is a part of a project of 120 acres, and the used irrigation system is pivot irrigation pumps. The main pumping source is the Nile River where 32 pumps are installed, each pump of 3200 L/s ($3.2 \text{ m}^3/\text{s}$) flowrate, 800 kW motor power and 613.2 kW rated power, Fig. 5 shows pictures of actual pumping system in the site under construction. The electrical power required by the pump (W_p) is a function of the pumped volume flow rate of water which is given as [30]:

$$W_p = \frac{Q\rho g(h + \Delta H)}{\eta_p \eta_m} \quad (6)$$

Where Q is volume flow rate (m^3/s), ρ is fluid density (kg/m^3), g is gravity (m/s^2), $(h + \Delta H)$ is the total dynamic head which is the summation of vertical suction lift (h) and friction losses (ΔH), η_p is the mechanical pump efficiency ranged from 90 to 95%, and η_m is the electric motor efficiency ranged between 45 and 65%.

The load profile of irrigation is considered constant as the consumption of the pumped water will be considered to remain the same for this study, the peak load is 1226 kW with energy consumption of 4385 KWh/day, the load is considered to cover two crops' circles, and this entails the use of two pivot irrigation pumps.

Sudan has abundance of solar irradiance with an average sunshine hour between 8.5 and 11 h per day. The solar energy density in Sudan is between 439 and 639 W/m^2 and annual global solar irradiation is over 2400 kWh/m^2 [31]. The annual daily solar radiation data needed for the study case location was provided by HOMER using the National Renewable Energy Laboratory Database and shown in Fig. 6. This data will provide the necessary information for the PV sizing and the amount of energy extracted by the solar PV



Fig. 4. Location of selected study case.



Fig. 5. Pumping system in the site.

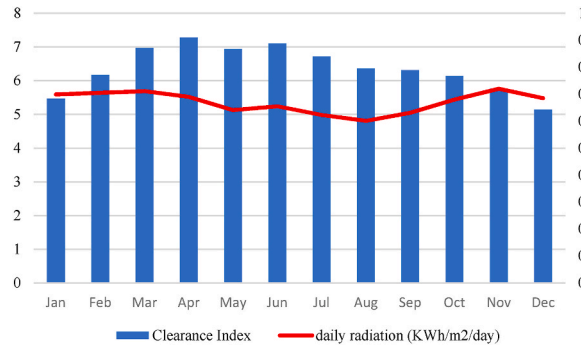


Fig. 6. Solar radiation data for Atbara, Sudan.

arrays.

4. Economical model

Economic aspect is one of the main factors that used to evaluate the feasibility of the any energy system is its cost. Thus, to ensure the hybrid energy system is cost-effective, all its components should be characterized by lowest total Net Present Cost (NPC) and Cost of Energy (COE) to be cost-effective.

$$NPC = \frac{C_{ann,tot}}{CRF_{(i,N)}} \quad (9)$$

Where $C_{ann,tot}$ is annual cost, i is real interest rate (6%), N is the lifetime of the hybrid system which is estimated to be “25 years”, and CRF is capital recovery factor which represented as follow:

$$CRF(i, N) = \frac{i(1+i)^N}{(1+i) - 1} \quad (10)$$

The total annual cost $C_{ann,tot}$ which include initial, maintenance and operation cost for all system components. In addition to replacement costs for battery storage.

The salvage value (SV) can be determined as follow:

$$SV = C_{rep} \frac{n_{rem}}{n_{comp}}, \quad (11)$$

Where C_{rep} is Components' replacement cost, n_{rem} and n_{comp} are the rest of the lifetime and the lifetime of the component respectively.

The COE can be examined as follow:

$$COE = \frac{C_{ann,tot}}{E_{prim} + E_{def} + E_{grid, sales}}, \quad (12)$$

Where E_{prim} , E_{def} and $E_{grid, sales}$ are the primary, deferrable and electrical energy sold to the utility grid.

5. Results and discussions

HOMER simulation is run for a hybrid system configuration, taking into consideration different energy management strategies and sizes of DG. Twelve alternatives that include LF, CC, combined LF – CC, and PC strategies of energy management and 500 kW, 750 kW, and 1000 kW diesel generators are investigated. During the assessment process, seven criteria that include various technical, economic, and environmental metrics are considered. The simulation is performed with an average diesel price in the world of 1.09\$/L. The simulation results for all twelve alternatives showed the optimum size of each case based on the minimum energy cost (COE). All the components for PV/DG/BB/FW/converter were simulated and optimized with the HOMER software. All dispatch control strategy methods were tested (LF, CC, CD and PS). The simulations for AC primary load adopted an hourly-based iterative process (8760 h). It commences with preliminary values to assess the capacity of energy components based on the required load. Thousands of feasible cases were repeated to specify the most reasonable size for the hybrid system components based on a minimum NPC and COE. The flowchart for control strategy of the proposed hybrid system is given in Fig. 7.

Table 3 presents the results of the twelve alternatives in terms of initial cost (\$), operating cost (\$/yr), net present cost (\$), COE (\$/kWh), renewable fraction (RF) (%), excess energy (%) and CO₂ emissions (kg/y). The COE values are arranged from 0.178 \$/kWh to 0.42 \$/kWh. Referring to Table 3, from an economic point of view, the minimum COE is 0.178 \$/kWh. It is achieved by alternative number eight that contains 1000 kW size of DG and applies combined LF–CC energy management strategy. But from an environmental point of view, the lowest CO₂ emissions is 8431 kg/y. It is achieved by alternative number four that contains 500 kW size of DG and applying load following energy management strategy. Therefore, it is challenging to directly determine the optimal energy management strategy and size of diesel generators because none of the twelve alternatives have the best parameters among the seven criteria. To overcome this contradiction, MCDM based no priority, CRITIC and Entropy weight approaches have been used to identify

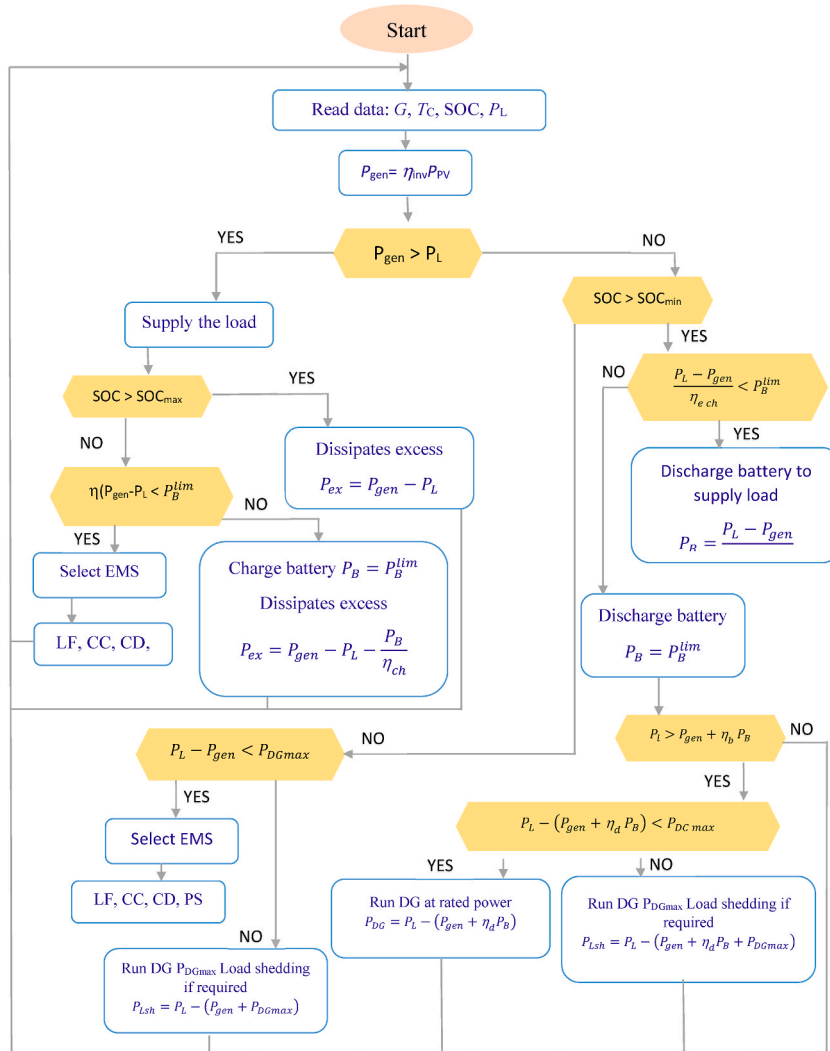


Fig. 7. Flowchart for control strategy.

Table 3
HOMER Pro results for twelve alternatives.

Alternatives	PV (kw)	FW	BS (Qty.)	Conv. (kw)	Initial Cost (\$)	Operating cost (\$/yr)	NPC (\$)	COE (\$/kwh)	RF (%)	Excess energy (%)	Emission (kg/y)
	Criterion				(C1)	(C2)	(C3)	(C4)	(C5)	(C6)	(C7)
Cycle charging											
A1	DG-500 kW	3017	1	456	1247	3.53 M	113856	5M	0.242	98.4	18470
A2	DG-750 kW	2164	1	408	1242	3.06 M	128458	4.72M	0.228	93.5	81742
A3	DG-1000 kW	2141	1	372	1242	3.01 M	125901	4.64	0.224	93.0	82351
Load following											
A4	DG-500 kW	3170	1	468	1283	3.65 M	112469	5.11 M	0.247	99.3	8431
A5	DG-750 kW	2197	1	396	1241	3.07 M	119581	4.61 M	0.223	95.1	63825
A6	DG-1000 kW	2410	1	384	1240	3.03 M	117678	4.55 M	0.22	95.1	60513
Combined LF-CC											
A7	DG-500 kW	3058	0	852	1033	3.54 M	138766	5.34 M	0.258	99.2	8795
A8	DG-750 kW	2703	0	588	1238	3.22 M	109910	4.64 M	0.224	99.0	11041
A9	DG-1000 kW	2072	0	216	1229	2.38 M	101526	3.69 M	0.178	91.0	98097
Predictive control											
A10	DG-500 kW	2425	0	576	1352	2.93 M	131918	4.64 M	0.224	94.9	57095
A11	DG-750 kW	1781	0	864	1362	3.01 M	438628	8.68 M	0.42	47.4	655216
A12	DG-1000 kW	1712	0	132	980	1.95 M	209841	4.66 M	0.225	73.5	322523

the best alternative. The three used methods of MCDM are TOPSIS, WASPAS and WSMWPM, and they are applied to decide the best alternative. In order to determine the importance of the criteria, no priority and CRITIC are used. The related values are presented in Table 4.

Table 5 and Fig. 8 show the final ranking based on the three MCDM methods (TOPSIS, WASPAS and, WSMWPM). As illustrated in Table 5, the final average rank for all alternatives has been determined. Alternative A8 which includes 750 kW DG size and combined LF-CC energy management is classified as the best option. Followed by alternative A4 which consists of 500 kW size of DG and applying LF management strategy. While A11 which, includes 750 kW DG size and predictive control energy management ranked as the worst alternative.

The analysis output of each technique along with different approaches are given in Fig. 8 (a, b, and c). Fig. 8 (a) shows the output of WASPAS technique, by using WASPAS technique A8 is the best alternative for the three different approaches No priority, CRITIC and entropy, followed by A4 then A7 for CRITIC and Entropy approaches while A11 is the worst alternative in the three different approaches. Fig. 8 (b) shows the output of WSMWPM technique, in these techniques also A8 is the best alternative followed by A4 for the three different approaches, then A7 for CRITIC and Entropy approaches while A11 is the worst alternative in the three different approaches. Fig. 8 (c) shows the output of TOPSIS technique, which is same as the output of WSMWPM techniques, where A8 is the best alternative followed by A4 for the three different approaches, then A7 for CRITIC and Entropy approaches, while A11 is the worst alternative in the three different approaches. The final ranking of the three techniques is given in Fig. 8 (d), which shows A8 as best alternative followed by A4, A7, A6, A1 and the worst one is A11.

The related cost and detailed performance of different components for the best and worst Alternatives A8 and A11 are shown in Table 6, Table 7 and Fig. 9 respectively. Comparing the total cost in Table 6 of each component of the two alternatives, showed that PV array has the highest cost among other components in the best alternative, followed by the BS. While in the worst option the cost of DG is the highest.

From Table 7 it is found that, the annual energy production (AEP) of PV array is equal to 5082 MWh in A8 with 0.0306 \$/kWh, while AEP for A11 is equal to 2495 MWh with 0.0411\$/kWh in A11. Thus, the AEP of A8 is almost twice the AEP of A11 and has less COE and capacity factor. By comparing the performance of DG in best and worst alternatives, it clear that the annual fuel energy input for A8 is 4,1397 KWh/y which is quite less than that generated by A11 which is equal to 2,456,576 KWh/y. However, this is expected as A11 has more annual operation hours (1758 h/year) less compared to A8. Accordingly, the total gas emission by A 8 is about 1.3% of the gas emission by A11, which is almost insignificant compared to A11.

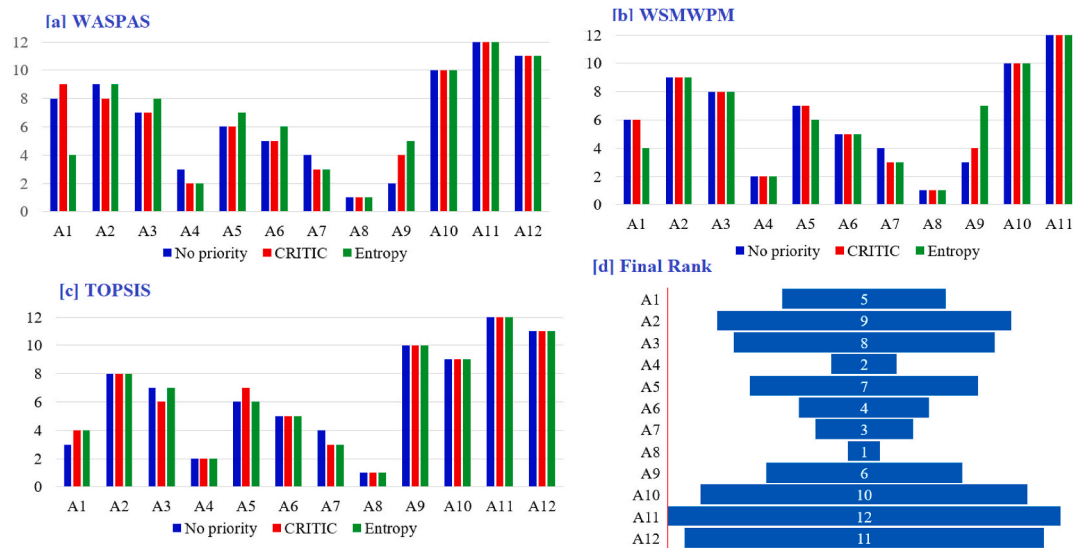
Table 4
Weights of criteria using No priority, CRITIC and Entropy.

Criterion		No priority	CRITIC	Entropy
Initial cost (\$)	C1	0.1429	0.1778	0.0375
Operating cost (\$/y)	C2	0.1429	0.0934	0.3815
Net present cost (\$)	C3	0.1429	0.0765	0.0718
COE (\$/kwh)	C4	0.1429	0.0765	0.0722
RF (%)	C5	0.1429	0.1108	0.0459
Excess energy (%)	C6	0.1429	0.2658	0.0201
Emissions (kg/y)	C7	0.1429	0.1993	0.371

Table 5

Ranking based on WASPAS, WSMWPM, and TOPSIS decision techniques.

	WASPAS						WSMWPM						TOPSIS						Final Rank
	No priority	Rank	CRITIC	Rank	Entropy	Rank	No priority	Rank	CRITIC	Rank	Entropy	Rank	No priority	Rank	CRITIC	Rank	Entropy	Rank	
A1	0.72018	8	0.67964	9	0.69966	4	0.71111	6	0.67104	6	0.68737	4	0.7497	3	0.6512	4	0.86186	4	5
A2	0.71393	9	0.68159	8	0.59758	9	0.69834	9	0.66251	9	0.57347	9	0.73163	8	0.61418	8	0.74341	8	9
A3	0.72134	7	0.68741	7	0.60483	8	0.7055	8	0.66803	8	0.57961	8	0.73788	7	0.61926	6	0.7459	7	8
A4	0.78555	3	0.77048	2	0.89406	2	0.77667	2	0.76062	2	0.88926	2	0.76369	2	0.69338	2	0.95194	2	2
A5	0.72733	6	0.68848	6	0.61938	7	0.71064	7	0.6686	7	0.59158	6	0.74023	6	0.61778	7	0.75121	6	7
A6	0.73567	5	0.69696	5	0.62577	6	0.71852	5	0.67636	5	0.5969	5	0.74573	5	0.62451	5	0.7528	5	4
A7	0.75557	4	0.75391	3	0.82313	3	0.74762	4	0.7451	3	0.81653	3	0.74591	4	0.69135	3	0.90036	3	3
A8	0.82635	1	0.80679	1	0.91764	1	0.82025	1	0.79983	1	0.91468	1	0.81405	1	0.74014	1	0.96495	1	1
A9	0.79649	2	0.72285	4	0.64801	5	0.76438	3	0.6863	4	0.58863	7	0.71247	10	0.54608	10	0.67518	10	6
A10	0.70763	10	0.66437	10	0.59226	10	0.69267	10	0.64727	10	0.56901	10	0.72902	9	0.59598	9	0.74028	9	10
A11	0.45667	12	0.51689	12	0.27735	12	0.42668	12	0.47352	12	0.26475	12	0.13369	12	0.23624	12	0.012254	12	12
A12	0.66334	11	0.65407	11	0.4293	11	0.62938	11	0.61029	11	0.39682	11	0.55351	11	0.42003	11	0.48489	11	11

**Fig. 8.** Results of MCDA ranking of the Twelve alternatives using solving approaches of: a) WASPAS, b) WSMWPM, c) TOPSIS, d) average ranking.**Table 6**

Related costs of different components for each considered system.

	Capital (\$)	Replacement (\$)	O&M (\$)	Fuel (\$)	Salvage (\$)	Total (\$)
Alternative A8 (Best option)						
DG	\$290,000.00	\$0.00	\$1,968.21	\$59,280.81	(\$57,793.33)	\$293,455.69
PV array	\$1,730,040.11	\$0.00	\$349,455.03	\$0.00	(\$69,074.11)	\$2,010,421.03
BS	\$705,600.00	\$534,340.83	\$456,082.78	\$0.00	(\$154,946.02)	\$1,541,077.59
Converter	\$495,191.11	\$0.00	\$320,079.56	\$0.00	(\$18,535.46)	\$796,735.21
System	\$3,220,831.22	\$534,340.83	\$1,127,585.59	\$59,280.81	(\$300,348.92)	\$4,641,689.53
Alternative A11 (Worst option)						
DG	\$290,000.00	\$247,769.72	\$164,767.66	\$3,517,844.73	(\$4,192.26)	\$4,216,189.84
PV array	\$1,139,996.33	\$0.00	\$230,270.65	\$0.00	(\$45,515.85)	\$1,324,751.14
BS	\$1,036,800.00	\$785,153.87	\$670,162.46	\$0.00	(\$227,675.78)	\$2,264,440.55
Converter	\$544,858.71	\$0.00	\$352,183.50	\$0.00	(\$20,394.56)	\$876,647.65
System	\$3,011,655.04	\$1,032,923.59	\$1,417,384.27	\$3,517,844.73	(\$297,778.45)	\$8,682,029.17

Table 7
The detailed performance of different component of the system.

Quantity	Units	Alternative A8 (Best option)	Alternative A11 (Worst option)
PV array			
Rated capacity	kW	2703	1781
Mean output	kW	580	285
Mean output	kWh/day	13925	6837
Capacity factor	%	21.5	16
Yearly energy production	MWh	5082	2495
PV penetration	%	318	156
hours of operation	hr/yr	4378	4382
Levelized cost	\$/kWh	0.0306	0.0411
DG			
Hours of operation	hr/yr	21	1758
Number of starts	Starts/yr	15	134
Operation life	yr	714	8.53
Capacity factor	%	0.24	13.3
Yearly energy production	kWh	15225	842357
Fuel consumption	L	4207	249652
Specific fuel consumption	L/kWh	0.28	0.296
Fuel energy input	KWh/yr	41397	2456576
Electrical efficiency	%	36.8	34.3
Battery storage			
Number of batteries	Qty.	588	864
Nominal capacity	kWh	4437	6520
Usable nominal capacity	kWh	2662	3912
autonomy	hr	14.6	21.4
Energy in	MWh/yr	650.91	411644
Energy output	MWh/yr	52.073	329923
Storage depletion	KWh/yr	121	-5841
Expected life	yr	12	12
Emissions			
Carbon Dioxide	kg/yr	8431	655216
Carbon Monoxide	kg/yr	43.6	2746
Unburned Hydrocarbons	kg/yr	2.31	315
Particulate Matter	kg/yr	0.373	78.6
Sulfur Dioxide	kg/yr	20.6	1635
Nitrogen Oxides	kg/yr	8.36	2746
Converter			
Capacity	kW	1238	1362
Hours of operation	hr/yr	1305	1305
Capacity factor	%	15.1	10.5
Losses	MWh/yr	83.433	66246

6. Conclusion

The attention towards the use of renewable energy in different life sectors are increasing rapidly in recent years for sustainable development, which includes energy security, clean environment, economic growth, and social development. One of these important sectors is agriculture which usually takes place in remote areas, where there is a lack of grid connections and high cost of diesel

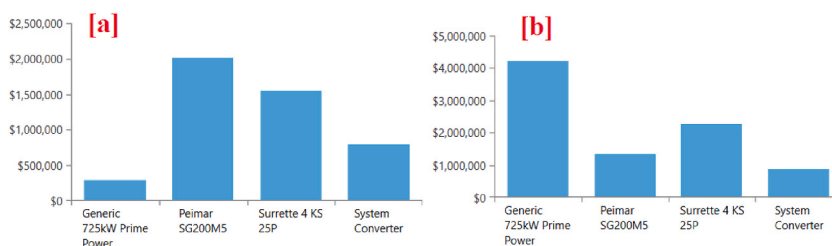


Fig. 9. The NPC for different system components [a] Alternative A8 (Best option) and [b] Alternative A11 (Worst option).

generators.

Techno-economic analyses of using hybrid energy systems for irrigation purposes in remote areas have been conducted in this study to evaluate its potential economically and environmentally. The system consists of PV, different sizes of DG, chemical, and mechanical storage systems, and twelve alternatives using those components have been proposed. The system power generation, energy cost, net present cost, fuel consumption, and carbon emission are all analysed, taking into consideration LF, CC, combined LF – CC, and predictive strategies energy management strategies.

From an economical point of view, minimum COE and NPC are the benchmarks to identify optimum options, accordingly, the best option is A9 which includes 1000 kW DG size/PV, and the chemical storage battery (BS) is the best since its COE and NPC are 0.178 \$/kWh and 3.69 M. However, and according to the environmental analysis, this selection is not accepted as it will produce 98097 kg/year of gas emission. Thus, and from the environmental point of view, the best option is option A4 which includes 500 kW DG size/PV, mechanical storage (FW), BS, as it produces 8431 kg/year of gas emission. However, the COE and NPC of this are 0.247\$/kWh and 5.11 M respectively which are extremely higher than A9.

To avoid such contradiction multi-criteria decision-making (MCDM) based on a different approach of weight is used. The selection by the MCDM excludes both A9 and A4 from the first ranking. It chooses A8 which includes 750 kW DG size/PV and chemical storage battery as the best option. The selection of MCDM takes into consideration both economic and environmental factors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- [1] J. Wang, T. Takarada, *ENVIRONMENTAL PROBLEMS ARISING from COAL HANDLING and PROCESSING*. Coal, Oil Shale, Natural Bitumen, Heavy Oil and Peat-Volume I, 2009, p. 273.
- [2] M. Kumar, Social, Economic, and Environmental Impacts of Renewable Energy Resources, Wind Solar Hybrid Renewable Energy System, 2020.
- [3] K. Obaideen, et al., On the contribution of solar energy to sustainable developments goals: case study on Mohammed bin Rashid Al Maktoum Solar Park, *International Journal of Thermofluids* 12 (2021), 100123.
- [4] D. Wichelns, J.D. Oster, Sustainable irrigation is necessary and achievable, but direct costs and environmental impacts can be substantial, *Agric. Water Manag.* 86 (2006) 114–127.
- [5] K. Shahverdi, et al., Solar-driven water pump with organic Rankine cycle for pressurized irrigation systems: a case study, *Therm. Sci. Eng. Prog.* 25 (2021), 100960.
- [6] H. Rezk, M. Abdelkareem, C. Ghenai, Performance Evaluation and Optimal Design of Stand-Alone Solar PV-Battery System for Irrigation in Isolated Regions: A Case Study in Al Minya (Egypt), vol. 36, *Sustainable Energy Technologies and Assessments*, 2019.
- [7] P. Purohit, Financial evaluation of renewable energy technologies for irrigation water pumping in India, *Energy Pol.* 35 (6) (2007) 3134–3144.
- [8] O.M. Ali, An experimental investigation of energy production with a hybrid photovoltaic/thermal collector system in Duhok city, *Case Stud. Therm. Eng.* 21 (2020), 100652.
- [9] C. Ghenai, et al., Design of hybrid solar photovoltaics/shrouded wind turbine power system for thermal pyrolysis of plastic waste, *Case Stud. Therm. Eng.* 22 (2020), 100773.
- [10] H. Jaber, et al., Effect of generator load on hybrid heat recovery system, *Case Stud. Therm. Eng.* 13 (2019), 100359.
- [11] W. Obaid, A.-K. Hamid, C. Ghenai, Hybrid water pumping system design: a case study in Dubai, United Arab Emirates, *Case Stud. Therm. Eng.* 26 (2021), 101121.
- [12] N. Khattab, et al., Feasibility of hybrid renewable energy water pumping system for a small farm in Egypt, *Int. J. Appl. Eng. Res.* 11 (11) (2016) 7406–7414.
- [13] S. Srinivasa Murthy, et al., Performance analysis of a stand-alone polygeneration microgrid, *Therm. Sci. Eng. Prog.* 19 (2020), 100623.
- [14] S. Srinivasa Murthy, et al., Parametric studies on a stand-alone polygeneration microgrid with battery storage, *Therm. Sci. Eng. Prog.* 19 (2020), 100608.
- [15] T. Salameh, et al., Integrated standalone hybrid solar PV, fuel cell and diesel generator power system for battery or supercapacitor storage systems in Khorfakkan, United Arab Emirates, *Int. J. Hydrogen Energy* 46 (8) (2021) 6014–6027.
- [16] B. Pollet, et al., Fuel-cell (hydrogen) electric hybrid vehicles, in: *Alternative Fuels and Advanced Vehicle Technologies for Improved Environmental Performance*, Elsevier, 2014, pp. 685–735.
- [17] A. Arabkoohsar, Chapter One - classification of energy storage systems, in: A. Arabkoohsar (Ed.), *Mechanical Energy Storage Technologies*, Academic Press, 2021, pp. 1–12.
- [18] M.E. Amiryar, K.R. Pullen, A review of flywheel energy storage system technologies and their applications, *Appl. Sci.* 7 (3) (2017) 286.

- [19] K. Pullen, A. Dhand, Mechanical and electrical flywheel hybrid technology to store energy in vehicles, in: *Alternative Fuels and Advanced Vehicle Technologies for Improved Environmental Performance*, Elsevier, 2014, pp. 476–504.
- [20] R. Sebastián, R.P. Alzola, Flywheel energy storage systems: review and simulation for an isolated wind power system, *Renew. Sustain. Energy Rev.* 16 (9) (2012) 6803–6813.
- [21] R. Arghandeh, M. Pipattanasomporn, S. Rahman, Flywheel energy storage systems for ride-through applications in a facility microgrid, *IEEE Trans. Smart Grid* 3 (4) (2012) 1955–1962.
- [22] R. Sebastián, R. Peña-Alzola, Control and simulation of a flywheel energy storage for a wind diesel power system, *Int. J. Electr. Power Energy Syst.* 64 (2015) 1049–1056.
- [23] M.A. Ramli, A. Hiendro, S. Twaha, Economic analysis of PV/diesel hybrid system with flywheel energy storage, *Renew. Energy* 78 (2015) 398–405.
- [24] M.S. Kyi, L. Maw, H.M. Tun, Study of Solar PV Sizing of Water Pumping System for Irrigation of asparagus, 2016, 5(06).
- [25] F. Diab, et al., An Environmentally-Friendly Tourist Village in Egypt Based on a Hybrid Renewable Energy System—Part One: What Is the Optimum City? 8 (7) (2015) 6926–6944.
- [26] D. Pavković, et al., Energy storage systems sizing study for a high-altitude wind energy application, *Energy* 76 (2014) 91–103.
- [27] M.A. Eltawil, Z. Zhengming, L. Yuan, A review of renewable energy technologies integrated with desalination systems, *Renew. Sustain. Energy Rev.* 13 (9) (2009) 2245–2262.
- [28] K. Brecl, Energy and Economic Yield of Photovoltaic Systems: reactive-Power Impact/Energijski in ekonomski donos fotonapetostnega sistema: vpliv jalove energije, *Elektrotehnikski Vestn.* 81 (1/2) (2014) 9.
- [29] V. Fthenakis, et al., Methodology Guidelines on Life Cycle Assessment of Photovoltaic Electricity, IEA PVPS Task, 2011, p. 12.
- [30] A. Barrueto Guzmán, et al., A Cost-Effective Methodology for Sizing Solar PV Systems for Existing Irrigation Facilities in Chile, vol. 11, 2018, p. 1853, 7.
- [31] M. El Zein, Solar Energy Potential in the Sudan, 2017.